Comparing X Sentiment with Pulse Asia's Rankings: The 8 Senators on the Edge of the Winning 12

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I. Introduction

The 2025 Philippine Elections are coming up in May, and it is a tight race for the 12 spots between the bottom 8 of Pulse Asia's list of 14 senators who have a chance to make the cut. The eight senators on the edge are the following: re-electionist Imee Marcos, former senator Ping Lacson, TV host Willie Revillame, re-electionist Ronald Dela Rosa, Makati Mayor Abby Binay, former senator Manny Pacquiao, House Deputy Speaker Camille Villar, and re-electionist Lito Lapid. These eight candidates are on the edge for a coveted seat in the Senate, and their standings in Pulse Asia's survey indicate that they are walking on a tightrope. Public perception, campaign efforts, and media narratives will play crucial roles in determining their final rankings. This article aims to compare the sentiment analysis of X discussions regarding these candidates with Pulse Asia's survey, and a Rappler article that references the survey to evaluate their standings to pinpoint contrasts between social media sentiments and mainstream media narratives.

II. Data Overview and Methodology

To analyze public sentiment, I used a dataset from Kaggle titled "Tweets on Philippine Elections 2025", which contains 21 thousand tweets in English, Tagalog, and Taglish related to the upcoming elections in May. I utilized Python for data preprocessing, sentiment classification, and visualization, leveraging libraries such as Pandas for data manipulation, Transformers for sentiment analysis, and Matplotlib/Seaborn for visualizations. Before running the dataset through a sentiment analysis pipeline, data preprocessing was conducted on the dataset, this included: dropping duplicate rows, dropping rows with null values in them, dropping unneeded columns, and normalizing text. Additionally, a new column named 'senator' was added to indicate the last name of the senator being talked about in the tweet. I then applied a transformer-based model named "RoBERTa", to classify tweets into positive, neutral, and negative sentiments and added those labels to a new column named 'sentiment', and only included entries that passed the 50% minimum confidence threshold. All of these procedures lead to a big trim from the original 21 thousand entries, to 486 entries. Although preprocessing was done on the dataset, there were numerous limitations and potential biases that could have influenced the confidence score of these sentiments, these will be discussed later on in the paper. These factors required manual quality assurance and a careful interpretation of analyzation results to ensure an acceptable assessment of social media sentiment.

III. Pulse Asia Survey and Rappler Article Insights

Pulse Asia recently released the results of their January 2025 Nationwide Survey on the upcoming May 2025 elections last February 10, 2025. They explicitly state that among the 66 senatorial candidates, 14 would have a statistical chance of winning if the May 2025 polls were held during the survey period. In that list of 14, it's the bottom 8 where things are getting tight, with only small percentage differences, at most under a point separating each contender from the next. It was mentioned that there were numerous key events before the survey was conducted that could have influenced the results, namely: survey include the the filing of impeachment complaints against Vice President Sara Duterte, a House megapanel recommendation for former president Rodrigo Duterte to face crimes against humanity charges over the war on drugs, and Marcos' decision to remove both Dutertes from the National Security Council, and lastly, the withdrawal of Francis Marcos from the senate race.

IV. Sentiment Analysis: Results and Key Insights

Table 2
2025 ELECTIONS: SENATORIAL PREFERENCES
January 18 - 25, 2025 / Philippines
(Multiple Responses, up to 12 names allowed)

page 1 of 2

Base: Total Interviews, 100%*

	Aware	Voting for	Rank
TULFO, ERWIN (LAKAS)	99	62.8	1
GO, BONG GO (PDPLBN)	99	50.4	2-3
SOTTO, TITO (NPC)	99	50.2	2-4
TULFO, BEN BITAG (IND)	97	46.2	3-8
CAYETANO, PIA (NP)	98	46.1	4-8
BONG REVILLA, RAMON, JR. (LAKAS)	98	46.0	4-8
MARCOS, IMEE R. (NP)	100	43.3	4-12
LACSON, PING (IND)	99	42.4	4-12
REVILLAME, WILLIE WIL (IND)	98	41.9	7-13
DELA ROSA, BATO (PDPLBN)	100	41.2	7-14
BINAY, ABBY (NPC)	96	41.1	7-14
PACQUIAO, MANNY PACMAN (PFP)	97	40.6	7-14
VILLAR, CAMILLE (NP)	93	38.4	9-14
LAPID, LITO (NPC)	99	37.7	10-14

Figure 1. Pulse Asia Survey Results

From Pulse Asia's survey, the rankings of the bottom 8 senators are as follows: Imee Marcos (7), Ping Lacson (8), Willie Revillame (9), Bato Dela Rosa (10), Abby Binay (11), Manny Pacquiao (12), Camille Villar (13), and finally, Lito Lapid (14).

According to AWS, sentiment analysis is defined as the process of analyzing digital text to determine if the emotional tone of the message is positive, negative, or neutral. Understanding political sentiment analysis is essential for grasping the dynamics and nuances of voter opinions during election cycles. By focusing on voter sentiment evaluation, analysts can decode public perceptions of candidates and policies (Williams, 2025).

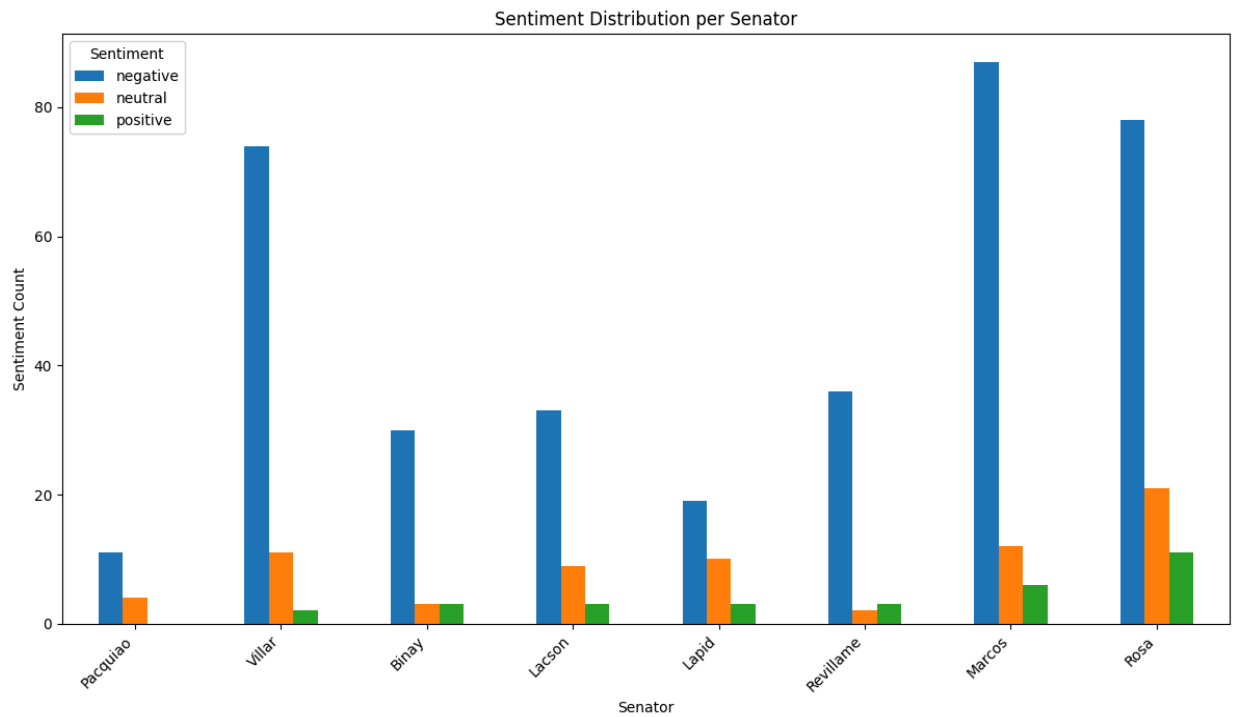

Figure 2. Grouped Bar Chart on the Sentiment Distribution Per Senator

This grouped bar chart represents the sentiment distribution of tweets mentioning specific senatorial candidates in the upcoming Philippine elections and is sorted from least to most positive sentiments. The candidates listed on the x-axis are the previously mentioned bottom 8 senators from the list of 14 that Pulse Asia has provided.

```
senator
Rosa         11
Marcos        6
Binay         3
Lacson        3
Lapid         3
Revillame     3
Villar        2
Pacquiao      0
Name: positive, dtype: int64
```

Figure 3. Count of Positive Sentiments Per Senator

From the visualization, trends and observations are the following:

1. There is a trend that all of the candidates have significantly higher counts of negative sentiments compared to neutral and positive sentiments.
2. Villar, Marcos, and Dela Rosa have received the highest number of negative sentiment tweets.

3. Dela Rosa has the highest number of positive sentiment tweets, Marcos comes in 2nd and Binay, Lacson, Lapid, and Revillame are all tied for 3rd.

Comparing the ranking of candidates from who has the least to most positive sentiments from this graph with the survey results from Pulse Asia, we can see that the rankings are different. From this grouped bar chart, we can see that the rankings (based on who has more positive sentiments) are as follows: Dela Rosa (7), Marcos (8), Lapid (T-9), Lacson (T-9), Revillame (T-9), Binay (T-9), Villar (13) and Pacquaio (14). Comparing these rankings with Pulse Asia's we can see that there is no similarity at all. The difference between these rankings from the ones from Pulse Asia suggests that online discourse does not directly align with survey-based voter preferences. Factors such as difference in data sources & methodologies, social media vs general voter base are all possible factors that could contribute to the discrepancy between the two rankings.

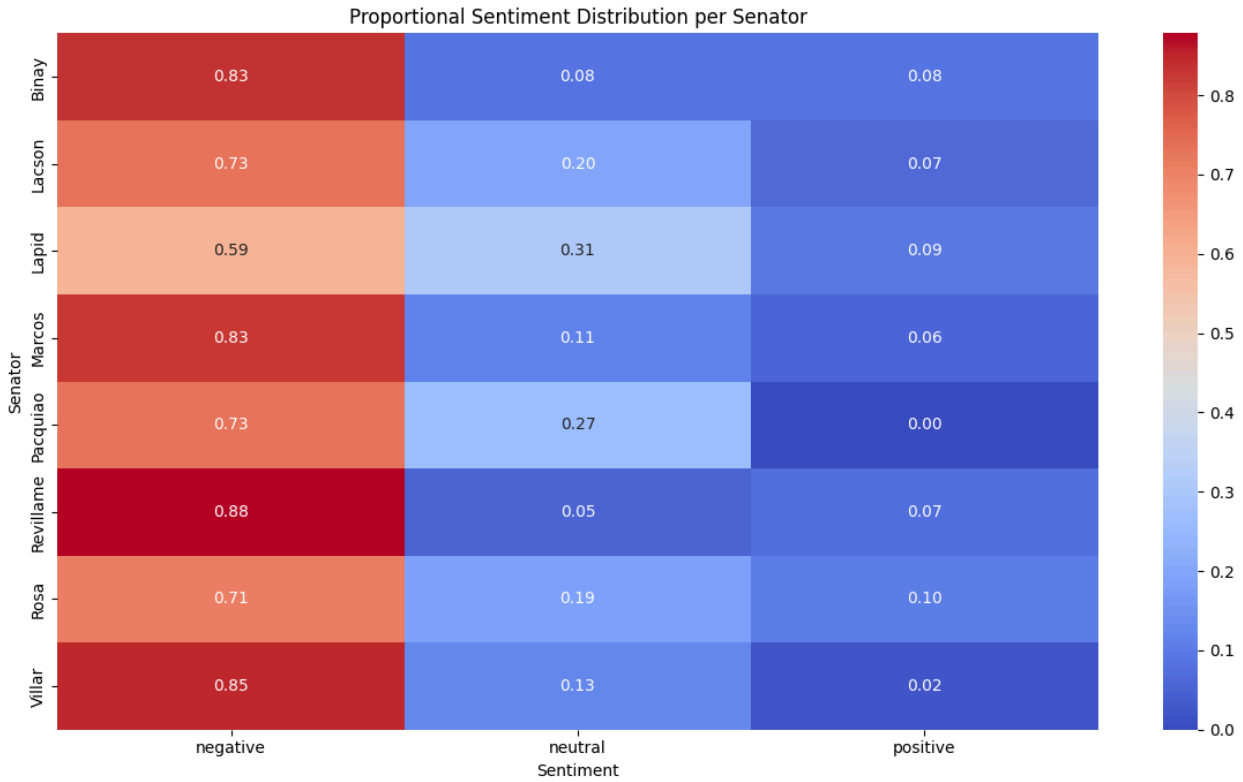

Figure 4. Heatmap on the Proportional Sentiment Distribution per Senator

This correlation heatmap visualizes the correlations between sentiment scores and the mentioned list of candidates. From the visualization, we can see that it follows the previous observation made through the bar chart, which is the idea that the trend of all candidates having a higher proportion of negative sentiments compared to the other two. Another interesting insight from this heat map is that the proportion of negative sentiments towards Binay, Marcos, Revillame, and Villar are all greater than 80%.

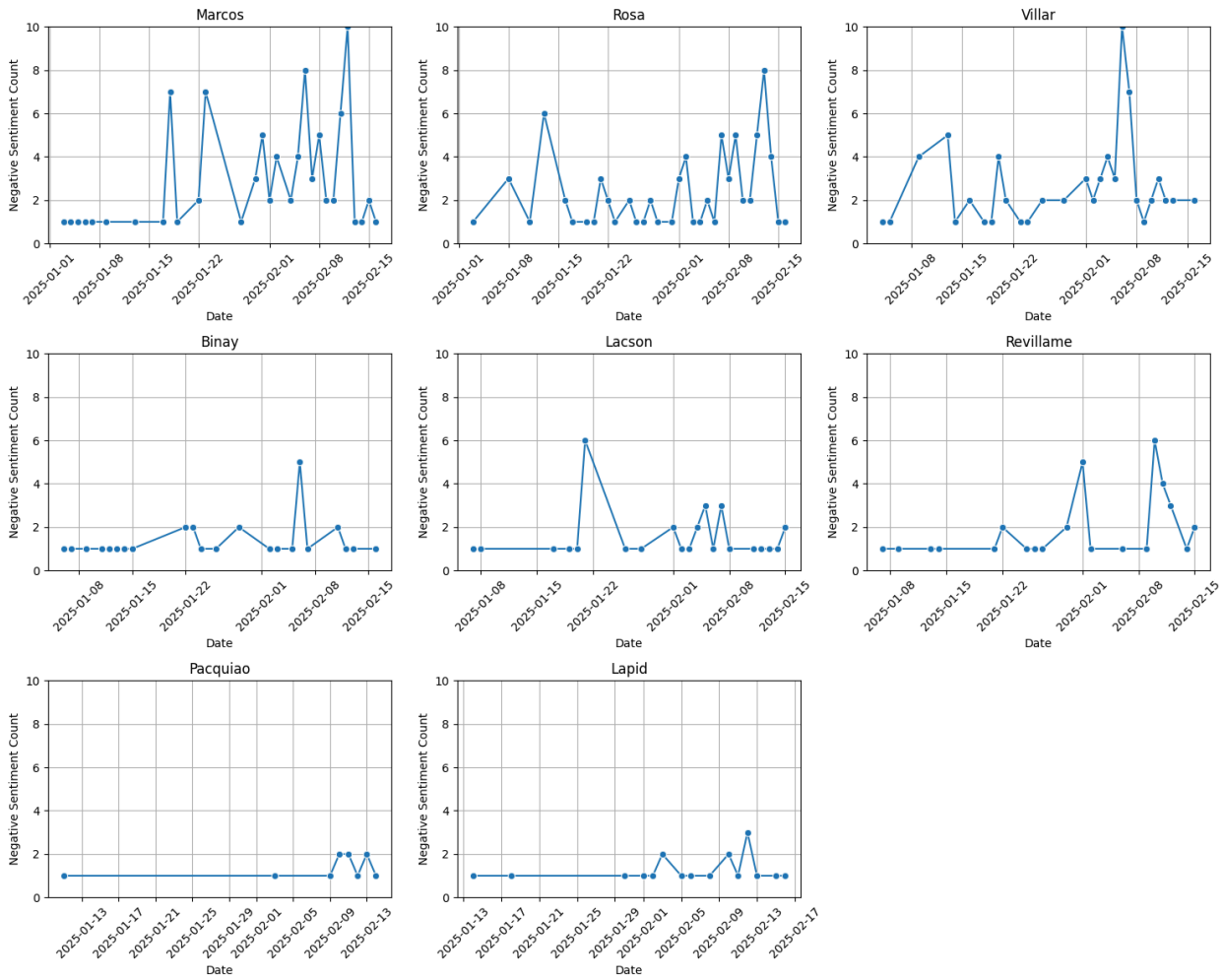

Figure 5. Time-Series Line Graphs on the Spikes and Dips in Negative Sentiment Over Time

These time-series line graphs aim to showcase the patterns and trends of negative sentiments toward senatorial candidates over time. From the graphs, we can see that the negative sentiments towards candidates on X fluctuates a lot and is volatile, especially in the cases of Marcos, Dela Rosa, and Villar. Since the negative sentiment count for those three is so high, it could be an indicator that they could be having specific controversies/events driving public discontent during those times.

V. Discussion of Analysis

The three visualizations demonstrated the volatile nature of social media discussions. While it shows that the candidates are actively discussed in social media, the sentiments toward them are mostly negative, giving the impression that public perception on social media platforms are skewed more towards criticism and dissatisfaction driven by current topics/events and immediate reactions rather than properly educated opinions formed over time. When comparing the sentiment analysis with Pulse Asia's survey results, this discrepancy can highlight the gap between volatile social media sentiments and stable public opinion captured through proper

survey and media reports. Such insights are crucial for the respective campaigns for these candidates, as it demonstrates the importance of addressing negative sentiment online and leveraging positive aspects to boost their public image to a wider demographic, boosting their chances to get elected.

Social media sentiment can indeed contribute to election outcomes, but in the case of predicting them, that is not the case. A previous research article by Macrohon et. al (2022) used sentiment analysis on tweets during the 2022 Philippine Presidential Election yielded some interesting results. Similar to my results, there was an overwhelming amount of tweets that were deemed to have a negative sentiment compared to the number of positive and neutral sentiment tweets in the data extracted. While tags relating to Leni Robredo dominated the dataset, particularly in terms of both positive and negative sentiments, it is clear she was a highly discussed figure. While she dominated both the positive and negative sentiments, it was also observed that tweets about her main opponent, Bongbong Marcos had more negative sentiments than positive sentiments by a huge margin. From these results, it would not be far-fetched to think that Leni would win the 2022 Presidential Elections but the actual election results demonstrated otherwise, as the number of votes in favor of Marcos overwhelmed Leni by a considerable margin. This is an example of the discrepancy between volatile social media sentiments and the broader public opinion.

VI. Limitations and Future Work

When working with this dataset there were 2 main limitations that could have influenced the results of the analyses. First is the sampling bias, it is important to note that X users may not be representative of the broader public sentiment, as it is only one of many platforms Filipinos can use to convey their sentiments. Second is language limitations, while the RoBERTa model used to perform sentiment analysis on the dataset is a model trained to handle multiple languages, it still doesn't undermine the fact that the Tagalog language has many nuances. Additionally, some tweets are in Taglish, which could further make it harder for the model to classify the sentiment of the tweets.

If an opportunity that would allow me to fine-tune and polish this paper were to present itself in the future, I would start by creating a structured workflow that expands the data sources to include other social media platforms such as Facebook, Reddit, and Threads. Additionally, I would implement two more machine learning techniques, Decision Tree and Random Forest for classifying sentiments. This approach would provide a more comprehensive analysis, enhancing the accuracy and depth of the sentiment classification.

VII. References

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VIII. Appendix

https://github.com/zookee76/robertaelections2025

Appendix A. GitHub Repository for Code